

# Sales Pulse

## Sales & Profit Analytics

CASE STUDY • END-TO-END BUSINESS INTELLIGENCE DASHBOARD

|                |               |           |
|----------------|---------------|-----------|
|                |               |           |
| <b>\$2.75M</b> | <b>\$776K</b> | <b>8K</b> |
| Total Revenue  | Total Profit  | Orders    |

### Project Focus

Revenue • Profitability • Product Performance • Customer Behavior • Geography

## 1. Business Problem & Objective

The business lacked a unified view of its sales performance across multiple countries (Egypt, Saudi Arabia, UAE, Jordan, Kuwait) and categories (Electronics, Fashion, Sports).

Management needed an interactive dashboard to track revenue, profitability, and customer behavior to make data-driven decisions on inventory and marketing spend.

### Goal:

Build an automated pipeline to clean raw data, transform it, and visualize key KPIs in a single source of truth.

## 2. Data Pipeline (The Tech Stack)

To ensure reliability, I built a complete ETL process:

### 1. Data Cleaning (Python):

- Used Pandas & NumPy to handle missing values, remove duplicates, and standardize date formats.
- Engineered new features like `Profit Margin` and `Average Order Value`.

### 2. ETL Automation (SSIS):

- Designed SQL Server Integration Services packages to automate the loading of cleaned data into a structured Data Warehouse.

### 3. Visualization (Power BI):

- Created interactive dashboards with DAX measures for dynamic filtering by year (2024-2026) and category.

## 3. Dashboard Walkthrough (Key Features)

The final dashboard consists of 5 interactive pages:

### Overview:

Top-level KPIs (Total Revenue: "\$2.75M", Profit: "\$776K", Orders: "8K").

**Product Performance:**

Identifies that "Electronics" (\$1.45M) is the top category, and "Dell Inspiron 14" (\$260K) is the best-selling product.

**Customer & Geography:**

Reveals that the Consumer segment drives 67% of sales, with "Cairo" and "Alexandria" being the top cities.

**Sales Channels:**

Shows that the "Marketplace" channel generates the highest revenue (\$1.51M).

**Insights Page:**

Translates numbers into actionable business recommendations.

## 4. Key Insights Discovered

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**Top Performer:**

Electronics drives 52% of total revenue, but Laptops specifically hold the highest margin.

**Geographic Focus:**

Egypt and Saudi Arabia together account for over 60% of total sales, indicating a strong regional concentration.

**Customer Behavior:**

The Consumer segment is massive (67%), but corporate clients have a higher Average Order Value (AOV) of \$450+.

**Seasonality:**

Sales peak in December and January, suggesting a holiday/New Year effect.

## 5. Business Recommendations

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Based on the analysis, I proposed the following actions:

- 1. Push High-Performing Products: Allocate more marketing budget to Dell Inspiron 14
- 2. Strengthen Electronics: Expand the Laptops subcategory by introducing complementary accessories.
- 3. Target High-Value Cities: Launch targeted retention campaigns in Cairo and Alexandria.
- 4. Balance Channels: While Marketplace is strong, the Mobile App is underperforming (only \$0.44M) – invest in App UX improvements.

## 6. Tools Used

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Python (Pandas) | SSIS | SQL Server | Power BI | DAX

## 7. Project Link

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[DoniaHabib1/salesPulse-analytics-dashboard](https://DoniaHabib1/salesPulse-analytics-dashboard):

End-to-end Sales Analytics project using SQL and Power BI to analyze revenue, profitability, product performance, customer behavior, and geographic trends.

